Laboratory I
Optical/X-ray Study of AGN

Joan Ribot Oliver

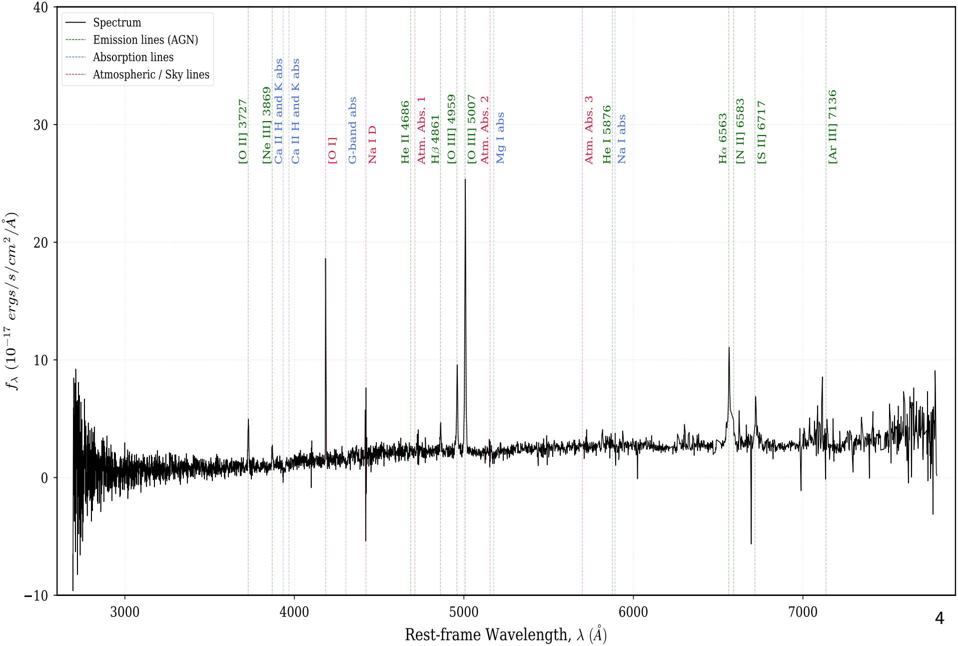
27/01/2026

Universidad de Cantabria

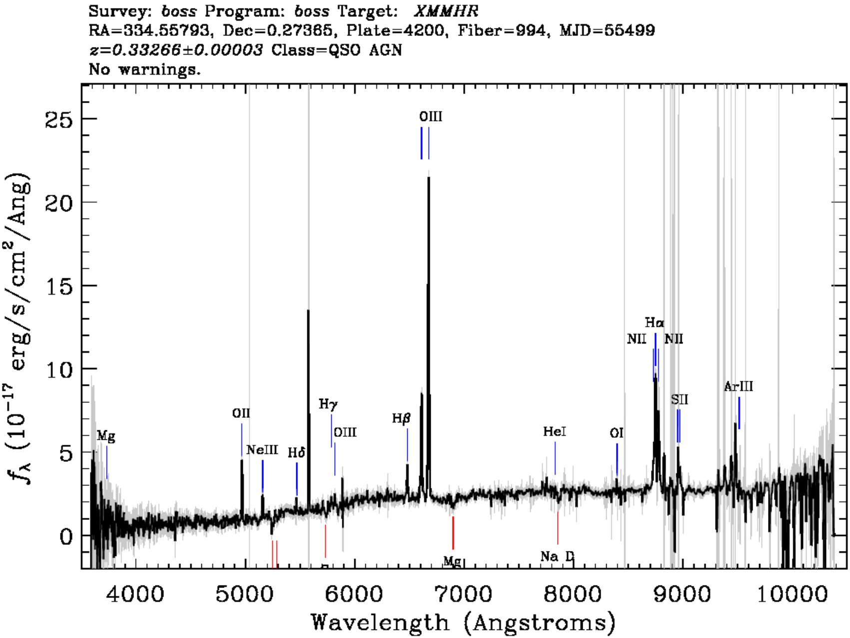
Index

SDSS ID	Plate	MJD	Fiber
	4200	55499	994
Coordinates	RA(°)	Dec(°)	
	333.165616	0.002465	

Table 1: SDSS identifier and coordinates of the studied AGN.


Spectrum
Emission lines (AGN)
Absorption lines
Atmospheric / Sky lines
[O II] 3727
[Ne III] 3869
Ca II H and K abs
Ca II H and K abs
[O I]
G-band abs
Na I D
He II 4686
Atm. Abs. 1
Hβ 4861
[O III] 4959
[O III] 5007
Atm. Abs. 2
Mg I abs
Atm. Abs. 3
He I 5876
Na I abs
Hα 6563
[N II] 6583
[S II] 6717
[Ar III] 7136
40
30
20
10
0
−10
3000
4000
5000
6000
7000
f_λ (10^{-17} $ergs/s/cm^2/\AA$)
Rest-frame Wavelength, λ ($\AA$)

1. Optical Spectra iii



- The redshift, z, of the studied AGN has been calculated in two different ways:
  1. First, using the initial approximation provided by the SDSS pipeline.
  2. Second, by identifying the main emission lines in the optical spectrum.
- For a flat ΛCDM Universe ($\Omega_m = 0.286$, $\Omega_\Lambda = 0.714$, $H_0 = 69.6$ km s^{-1} Mpc^{-1}), the following distances have been calculated using both redshift values using Ned Wright's Cosmology Calculator.[1]

	z	D_A (Mpc)	D_L (Mpc)	Proper distance (Mpc)
First approximation	0.33266 ± 0.00003	994.4	1766.1	1325.3
Mean from lines	0.3328 ± 0.0005	994.8	1767.3	1325.9

Table 2: Redshift and distances of the studied AGN, first with an approximation from the pipeline and then the mean from identified lines.

[1] `http://www.astro.ucla.edu/~wright/CosmoCalc.html`

1.1. Determination of redshift and distances ii

- The first aproximation of the redshift provided by the '*Z*' and '*Z_ERR*' columns of the SDSS pipeline analysing the *spec-2.fits* file.

- To determine the redshift more accurately, the main emission and absorption lines in the optical spectrum are identified and fitted, and then it is calculated for each line with the following equation:

$$z = \frac{\lambda_{obs} - \lambda_{theo}}{\lambda_{theo}} \tag{1}$$

  where λ_{obs} is the wavelength observed in the spectrum, and λ_{theo} is the theoretical rest-frame wavelength of the line. The mean value from all these measurements is the one shown in Table 2.

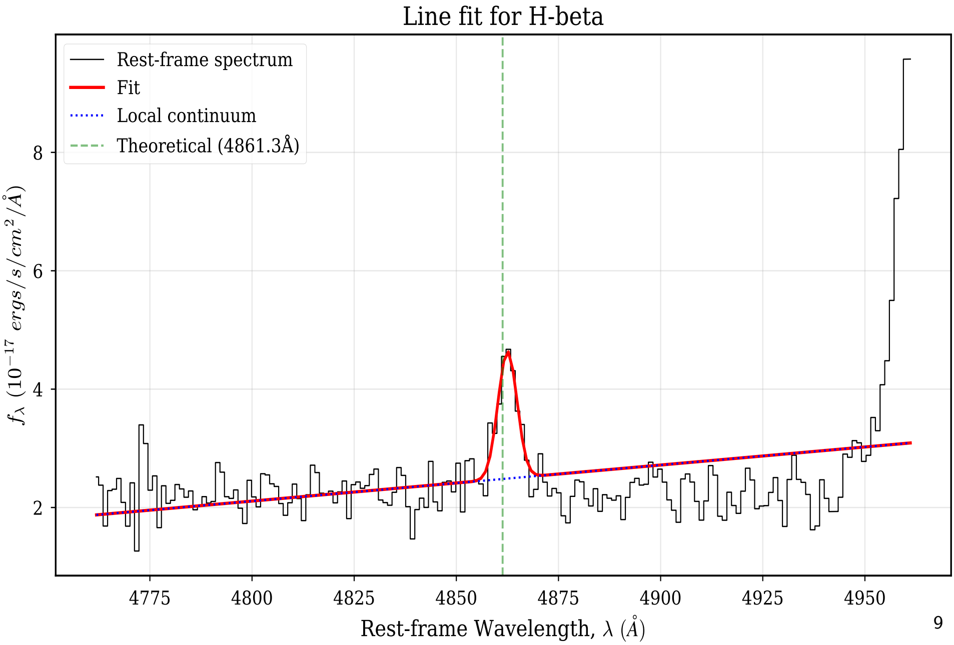
- The absorption and emission lines used are listed in Tabla 1 of the lab guide, with some extra lines.

1.2. Spectral fit i

- The centre, λ_{obs}, and width, FWHM, of the main emission lines in the optical spectrum of the studied AGN have been determined by fitting Gaussian profiles to the lines.

- An example of the fit of the Hβ emission line is shown in the following frame.

1.2. Spectral fit ii



1.2. Spectral fit iii

- From the fitted Gaussian profiles, the velocity dispersion, σ_v, of the gas emitting each line has been calculated using the relation:

$$\sigma_v = \frac{\text{FWHM}}{2\sqrt{2\ln(2)}} \approx 0.4247 \times \text{FWHM} \tag{2}$$

where FWHM is the Full Width at Half Maximum given in km/s.

Emission Line	λ_{theo} (Å)	λ_{obs} (Å)	FWHM (Å)	σ_v (km/s)
[O II]	3727.1	3728.4 ± 0.15	6.6 ± 0.4	530 ± 30
[Ne III]	3868.8	3869.9 ± 0.4	7.1 ± 0.9	550 ± 70
Hβ	4861.3	4862.5 ± 0.8	5.5 ± 1.9	338 ± 120
[O III]	5006.8	5008.3 ± 0.1	7.1 ± 0.3	425 ± 16
Hα	6562.8	6568.3 ± 1.7	37 ± 4	1700 ± 200
[N II]	6583.5	6568.2 ± 1.8	36 ± 5	1600 ± 200
[S II]	6716.4	6724.7 ± 3.0	20 ± 7	900 ± 300

Table 3: Theoretical, λ_{theo}, and observed, λ_{obs}, wavelength, FWHM and σ_v of the main emission lines in the optical spectrum of the studied AGN.

1.3. Black Hole Masses i

- In general, the lines are narrow (FWHM < 1000 km/s), typical of type 2 AGN.
- However, at higher wavelengths some lines such as the Hα's one appears significantly broader (FWHM ~ 1700 km/s).
- This suggests that the studied AGN could be a type 1.8-1.9 AGN, where some broad line region (BLR) emission is visible.
- The exctinction affects bluer wavelengths first, so the Hβ line appears narrow while Hα is broader.
- This can be explained because the radiation recieved is not the primary one, but the one that has been scattered and reflected by material surrounding the AGN.

- The mass of the SMBH can be calculated as follows:

$$M_{BH}(H_\beta) = 1.05 \times 10^8 \left(\frac{L_{5100}}{10^{46}\ \mathrm{erg/s}} \right)^{0.65} \left[\frac{FWHM(H_\beta)}{10^3\ \mathrm{km/s}} \right]^2 M_\odot, \quad (3)$$

$$M_{BH}(MgII) = 5.6 \times 10^6 \left(\frac{L_{3000}}{10^{44}\ \mathrm{erg/s}} \right)^{0.62} \left[\frac{FWHM(MgII)}{10^3\ \mathrm{km/s}} \right]^2 M_\odot. \quad (4)$$

  although in this case MgII is hard to distinguish and Hβ is narrow, (338 $\pm$ 120 km/s).

- Therefore, the luminosities L_{5100} and L_{3000} cannot be accurately determined, and the mass estimation is not carried out.

2. X-ray spectra i

- The X-ray spectrum of the studied AGN has been fitted using HEASoft/Xspec[2] software in Python[3] and following the Peter Boorman's tutorials 0 & 1.[4]

- The hydrogen column density, N_H, and the photon index, Γ, have been determined from the fit, and the statistical significance of each model has been evaluated using Bayesian statistics and the Akaike Information Criterion (AIC).

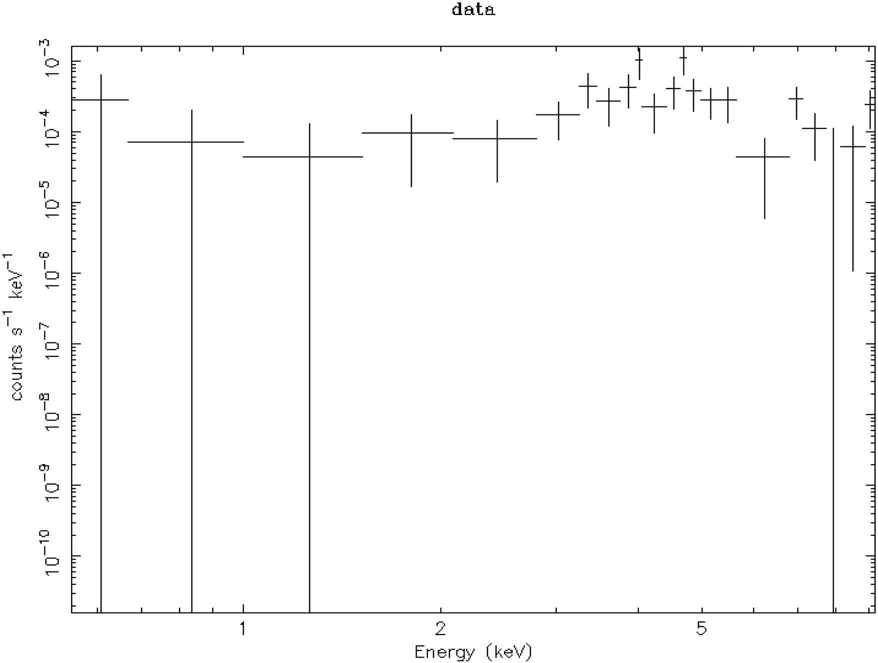
---

[2] https://heasarc.gsfc.nasa.gov/docs/software/xspec/

[3] https://www.python.org

[4] https://peterboorman.com/tutorial_bxa.html#welcome

2.1. Inspect and plot the data



- The X-ray spectrum has been fitted using five different models from simple to complex, as listed in the following table:

Number	Name	Code
1	Simple powerlaw	zpowerlw
2	Abs. powerlaw	TBabs*zTBabs*(zpowerlw)
3	Abs. powerlaw + Gaussian	TBabs*zTBabs*(zpowerlw+zgauss)
4	Abs. pl. + Gaus. + Compton Scat.	TBabs*zTBabs*cabs*(zpowerlw+zgauss)
5	Double Abs. powerlaw + Gaus.	TBabs*(zpowerlw+zTBabs*cabs*(zpowerlw+zgauss))

Table 4: X-ray spectral models and their corresponding codes used in the fitting process.

- The results of the Bayesian statistics and the AIC values for each model are shown in the next table:

Model	logZ	PhoIndex (Γ)	log(norm)	$\log(N_H)$	AIC
1	-57.23 ± 0.18	1.03 ± 0.03	-5.63 ± 0.07	–	104.38
2	-20.60 ± 0.20	1.91 ± 0.15	-4.10 ± 0.18	1.61 ± 0.13	33.34
3	-42.8 ± 0.4	1.95 ± 0.14	-3.80 ± 0.14	1.97 ± 0.03	72.14
4	-31.6 ± 0.3	1.96 ± 0.15	-3.50 ± 0.17	1.95 ± 0.04	50.61
5	-20.50 ± 0.25	1.7 ± 0.3	-4.18 ± 0.33	1.58 ± 0.14	33.04

Table 5: Bayesian statistics results and AIC values for all the models fitted.

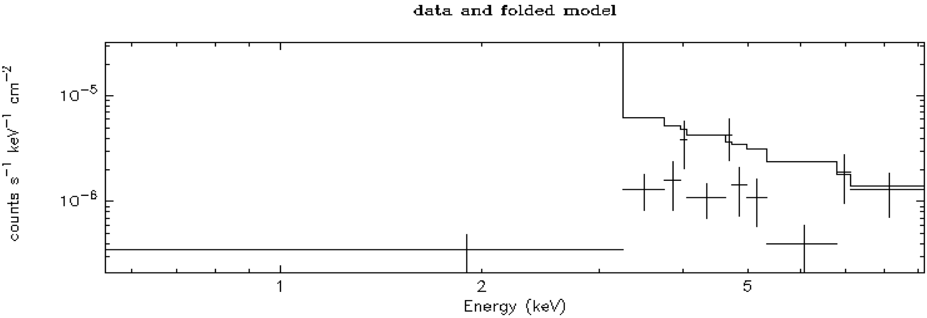
2.2. Fit the data iii

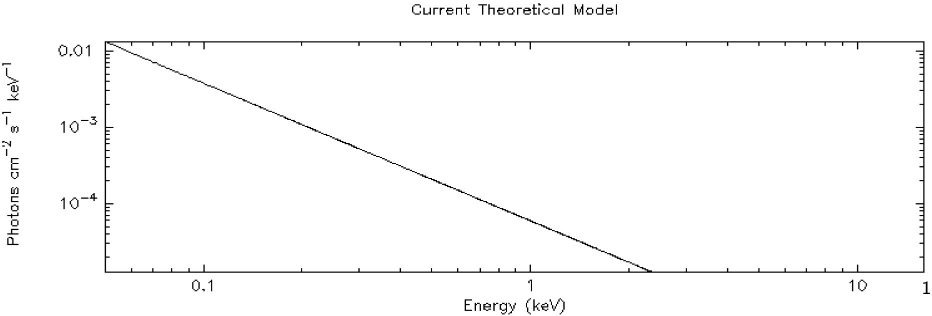
- The comparison of the models fitted to the X-ray spectrum with the log10(Z) method is carried out with the *model_compare.py* script provided in Peter Boorman's tutorial.[5]

```
Model comparison
****************

model powerlaw_pyxspec/: log10(Z) =   -16.5  XXX ruled out
model tbabspowerlaw+gaussian_pyxspec/: log10(Z) =    -9.8  XXX ruled out
model tbabspowerlaw+gaussian+cabs_pyxspec/: log10(Z) =    -4.8  XXX ruled out
model tbabspowerlaw_pyxspec/: log10(Z) =    -0.1    <-- GOOD
model double_powerlaw_pyxspec/: log10(Z) =     0.0    <-- GOOD

The last, most likely model was used as normalization.
Uniform model priors are assumed, with a cut of log10(30) to rule out models.
```

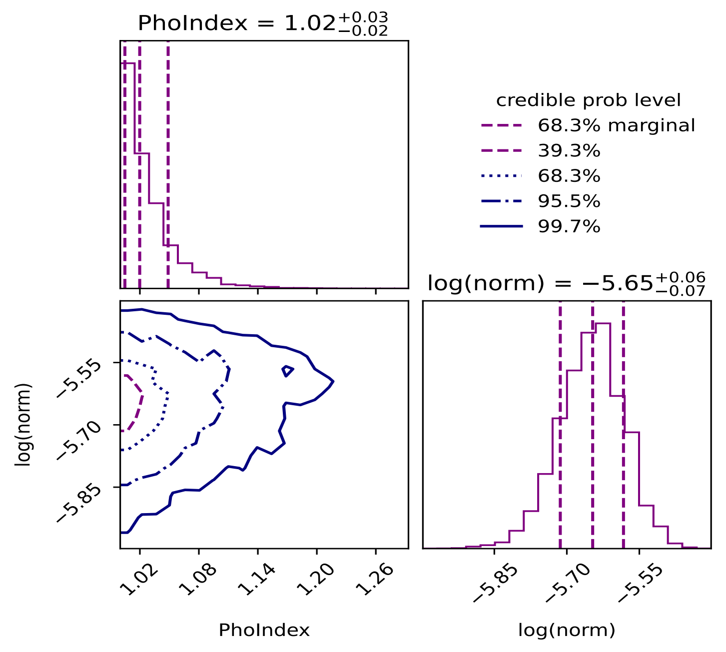
[5] https://peterboorman.com/tutorial/session3.html

2.2. Fit the data iv

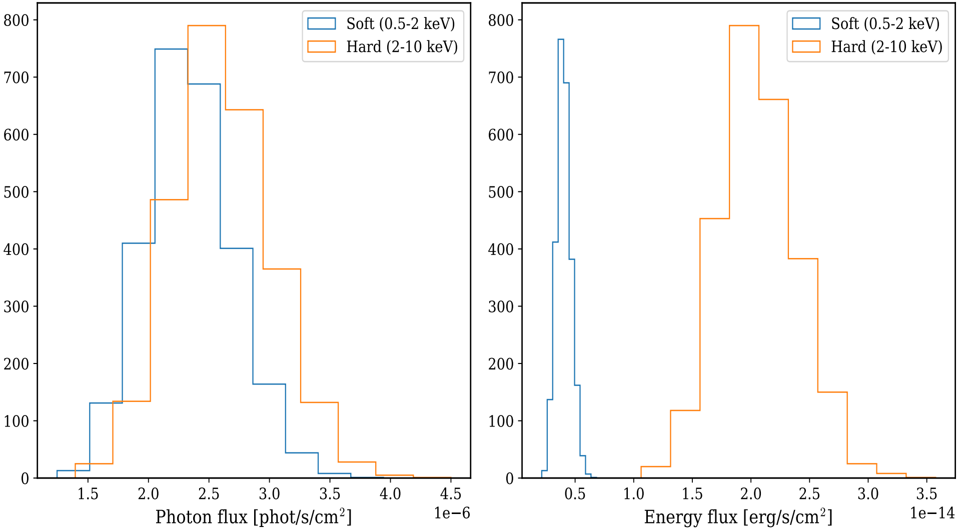




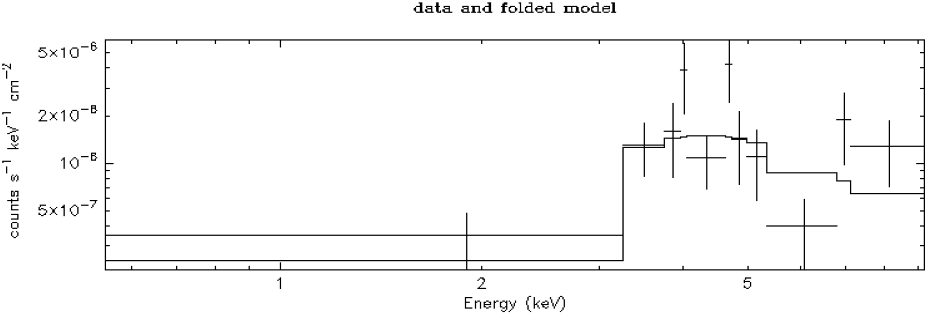
2.2. Fit the data v

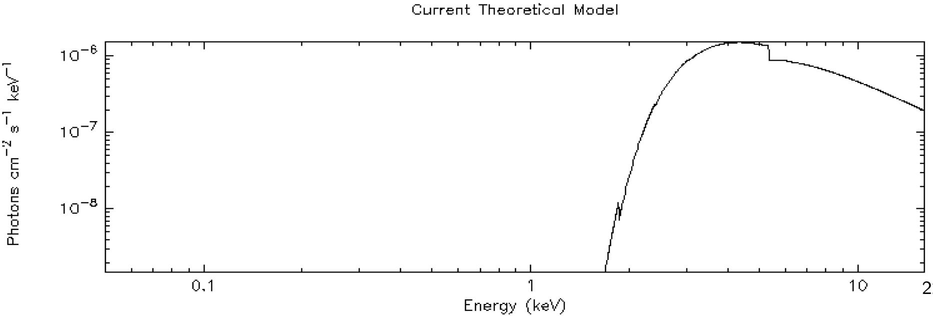


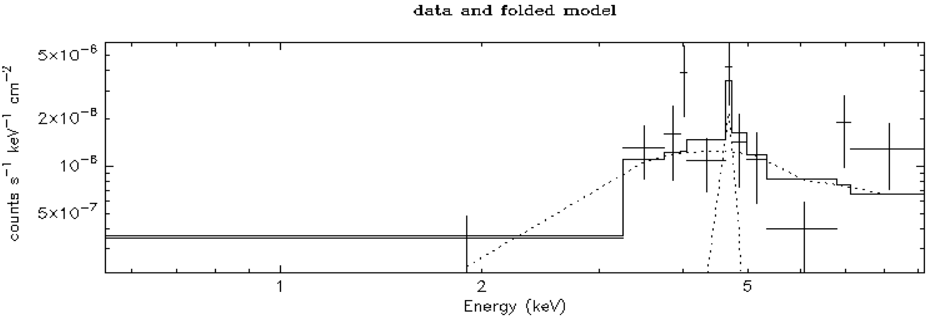
2.2. Fit the data vi

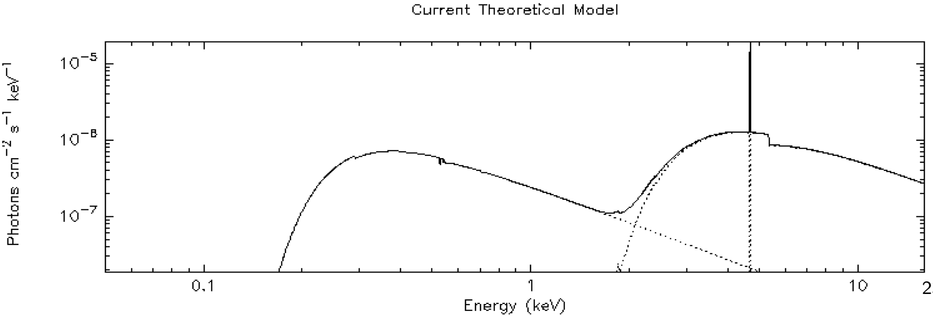


2.2. Fit the data vii






data and folded model
counts s^{-1} keV^{-1} cm^{-2}
5×10^{-6}
2×10^{-6}
10^{-6}
5×10^{-7}
1
2
5
Energy (keV)


Current Theoretical Model
Photons cm^{-2} s^{-1} keV^{-1}
10^{-5}
10^{-6}
10^{-7}
0.1
1
10
Energy (keV)

3. Results

- Consistency between the results from the Optical and the X-ray spectra.
- Okham's razor for model selection in X-ray spectra suggests that the best model is the absorbed power-law (model 2) instead of the double absorbed power-law (model 5).
- Both models suggest that the AGN is heavily obscured ($N_H > 10^{22}$ cm^{-2}) and has a photon index $\Gamma \sim 1.9$, typical values for Type 2 AGNs.
- However, the width of some emission lines in the optical spectrum, such as Hα (FWHM $\sim$ 1700 km/s), suggests that the AGN could be a Type 1.8-1.9 AGN.
- The mass of the SMBH could not be estimated due to the difficulty in accurately determining the luminosities L_{5100} and L_{3000}, as the Hβ line is narrow and the MgII line is hard to distinguish.

Thanks for your attention!

Round of questions.

Unified Model of AGN